signal and noise

prof. dr. nicolas meseth

1. what the numbers say
2. what decides your alphabet
3. build evidence

part 1

what the numbers say

the lamp is as bright as ever, so why does its beam end so soon?

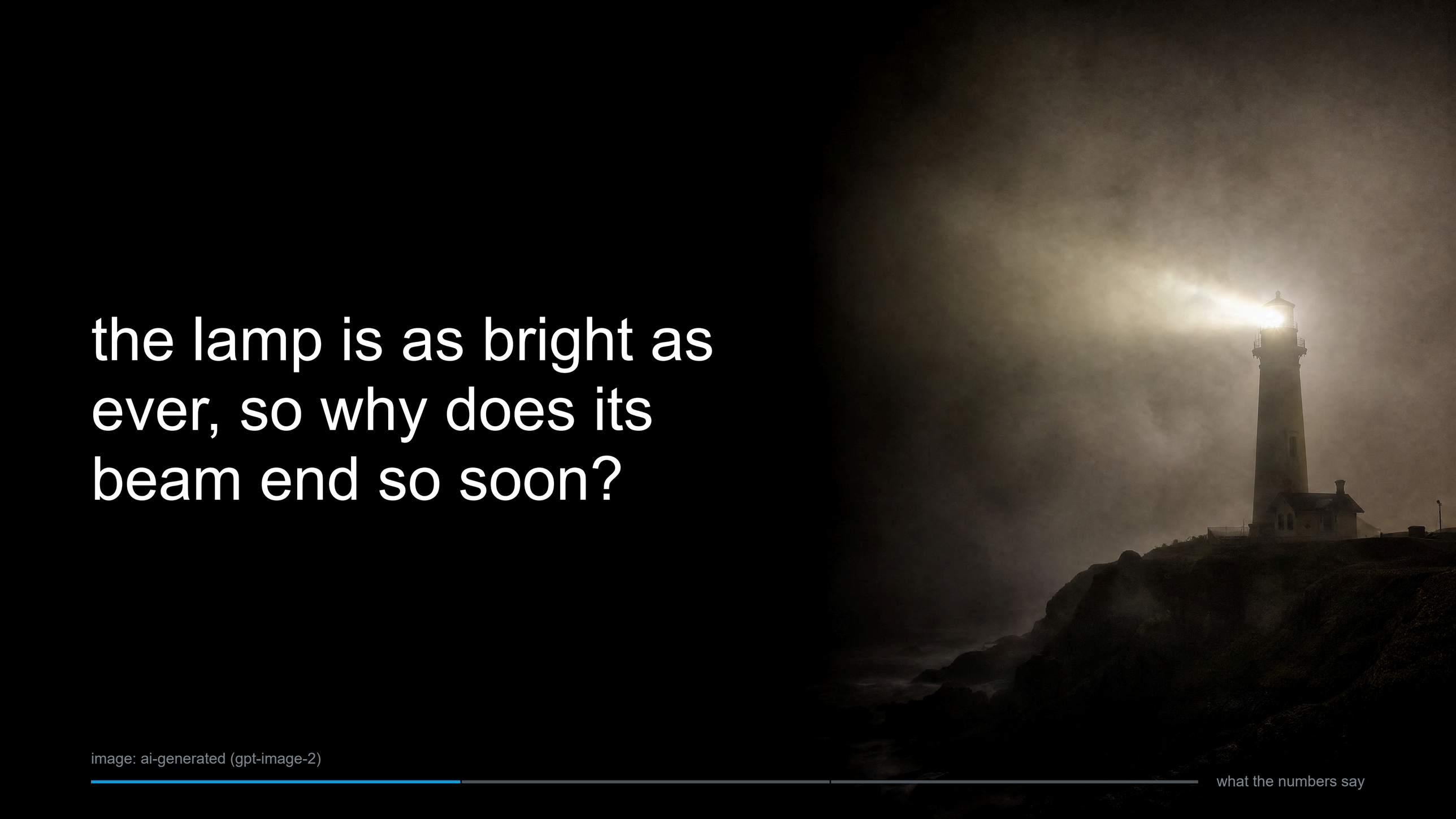


image: ai-generated (gpt-image-2)

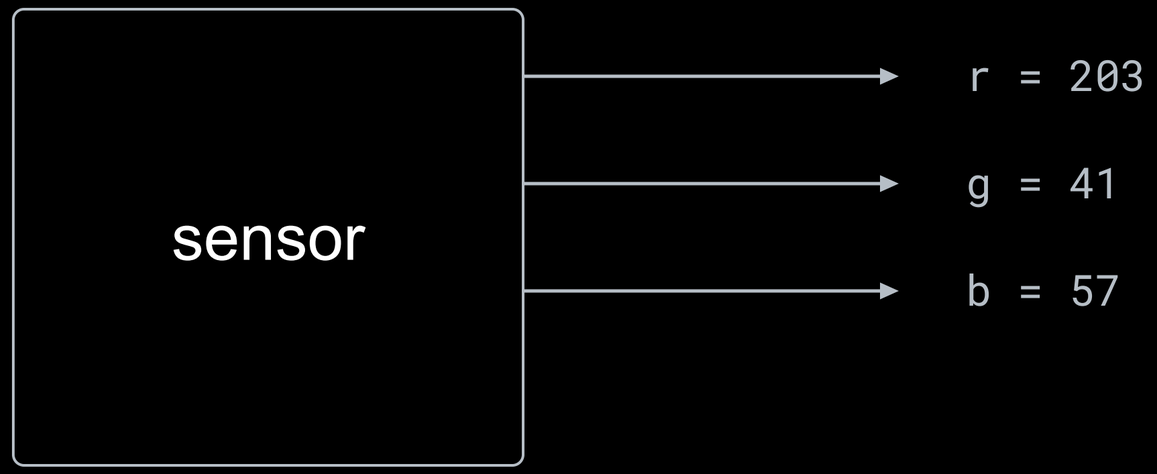
same hardware, different alphabets

team	alphabet	hit rate	what they did about noise
team a	16	98 %	cardboard tube, fixed distance, mean of 8
team b	8	96 %	longer integration time, shielded from the window
team c	8	91 %	picked colours further apart
team d	4	100 %	nothing, played it safe
team e	4	82 %	nothing, and it still failed

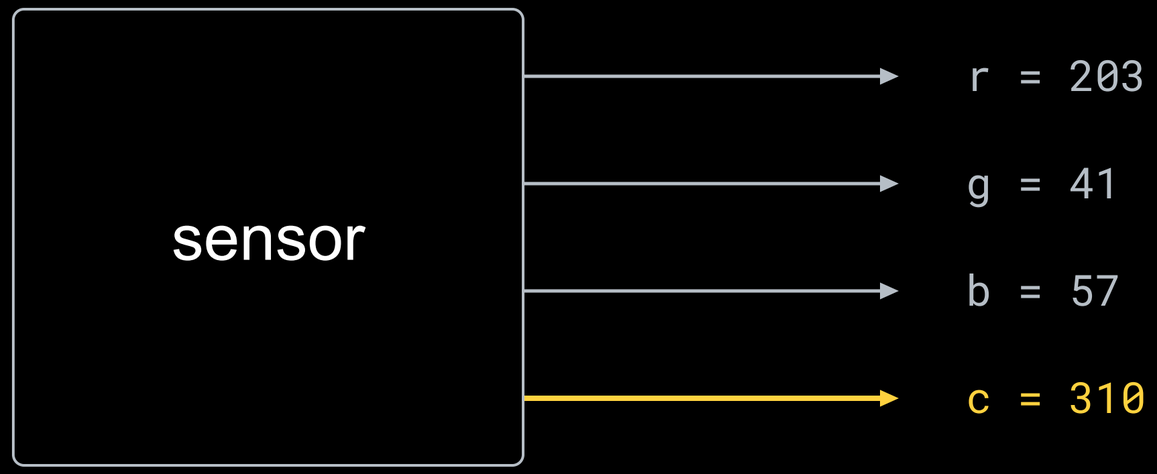
with the same hardware and task, the alphabets still differ by a factor of four.

example values; add your own team as a sixth row

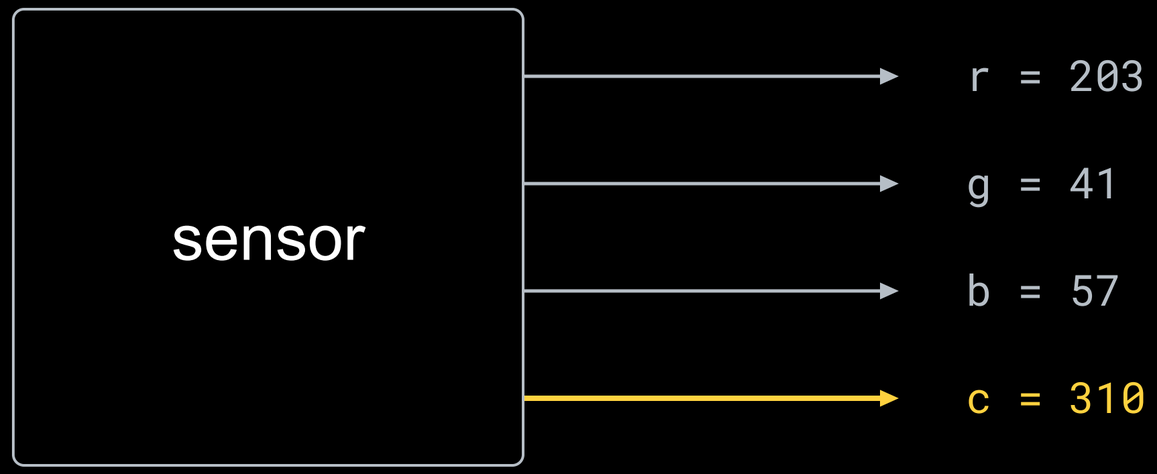
the sensor returns four numbers



the sensor returns four numbers



the sensor returns four numbers



`clear`

no colour filter

most sensitive

your eyes see a colour, but the program only sees four numbers.

sent is not measured

what you send

```
r = 255
g =   0
b =   0
```

sent is not measured

what you send

```
r = 255
g =   0
b =   0
```


what arrives

```
r = 203
g =  41
b =  57
c = 310
```

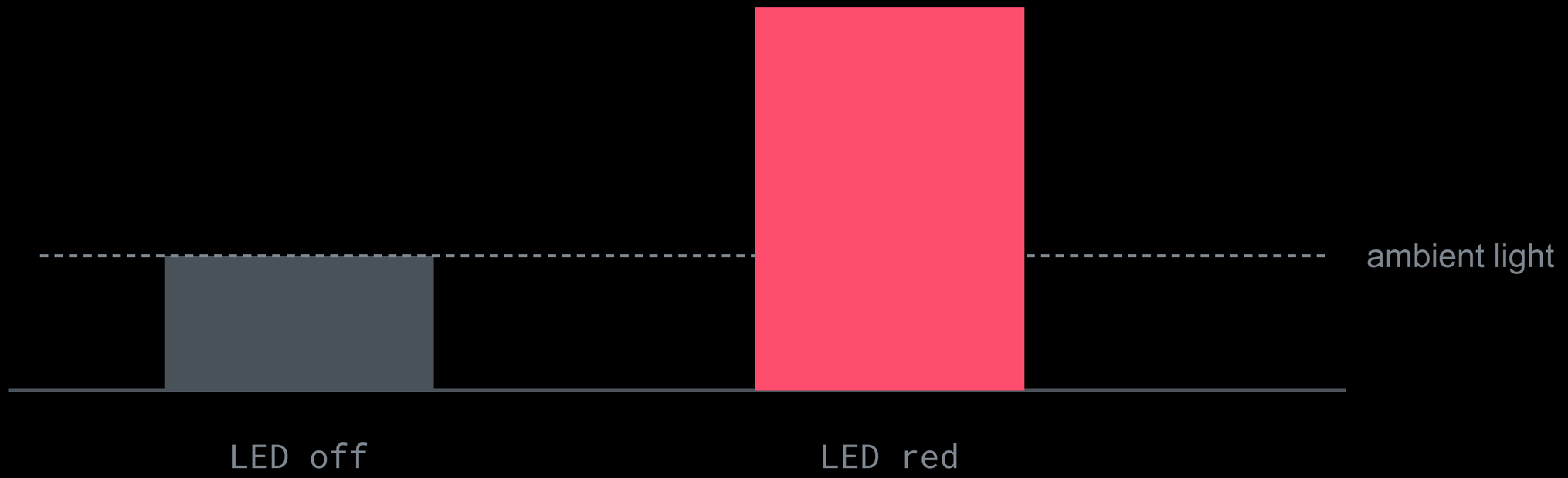
the filters overlap, distance dims the light, and the room shines in too

the numbers differ from what was sent, and that is normal, not a defect.

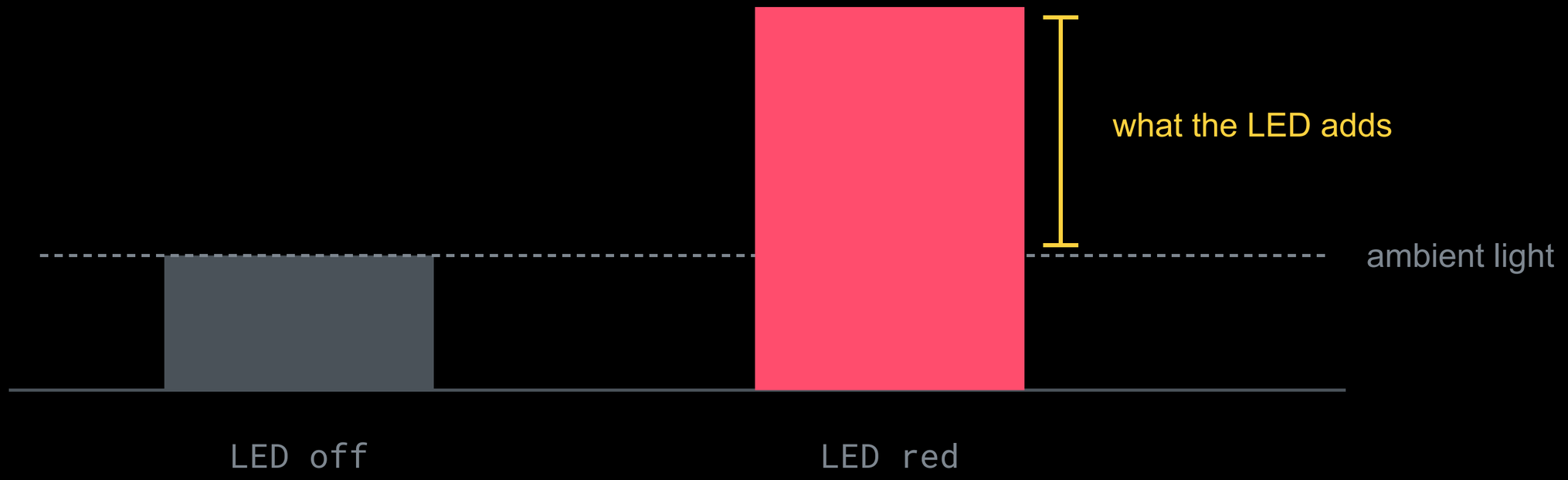
measure the dark first



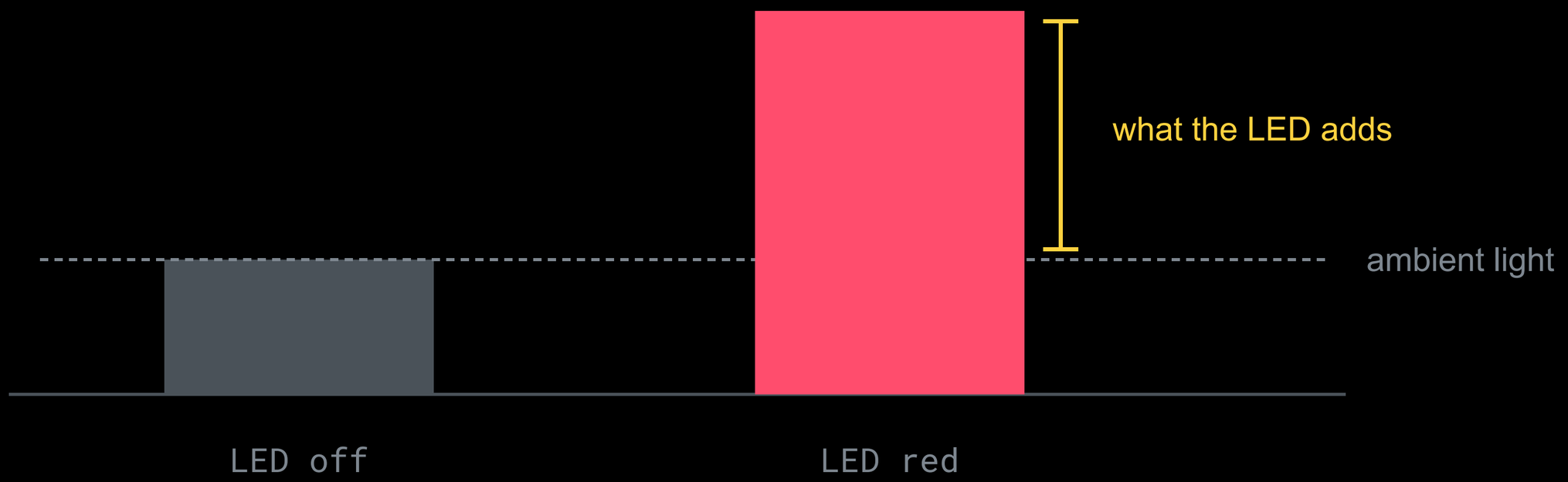
measure the dark first



measure the dark first

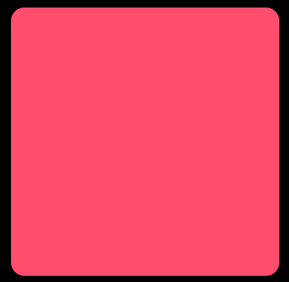


measure the dark first




```
reading = ambient + LED + noise
```

ten readings of one colour

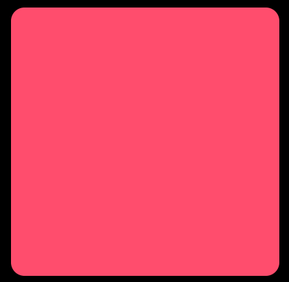


LED red
ten readings

what the sensor reports

214

ten readings of one colour



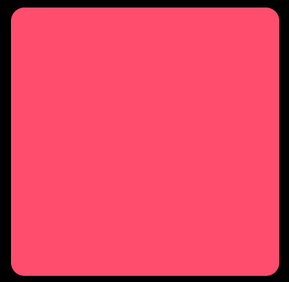
LED red
ten readings

what the sensor reports

214 211 218 209 216 213 220 207 215 212

nothing changed between the ten readings, and still every number is different.

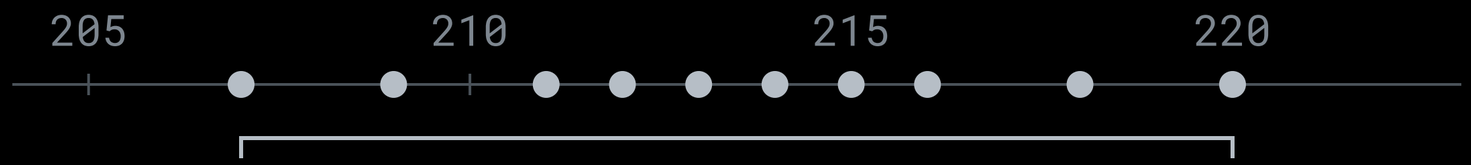
ten readings of one colour



LED red
ten readings

what the sensor reports

214 211 218 209 216 213 220 207 215 212

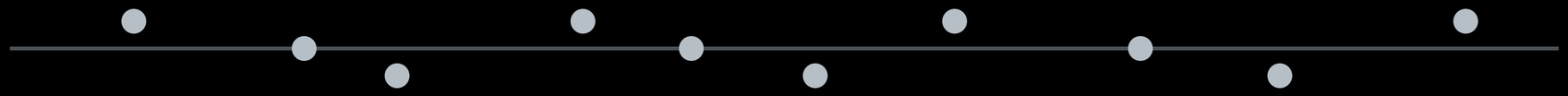


spread: 207 to 220

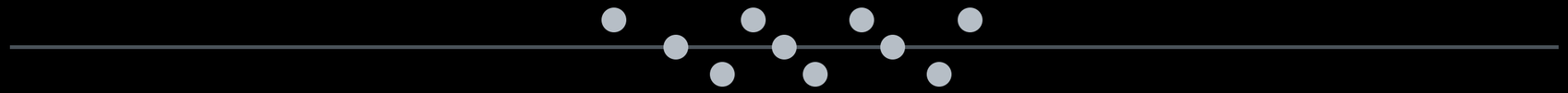
nothing changed between the ten readings, and still every number is different.

a series reveals what one reading hides

individual readings

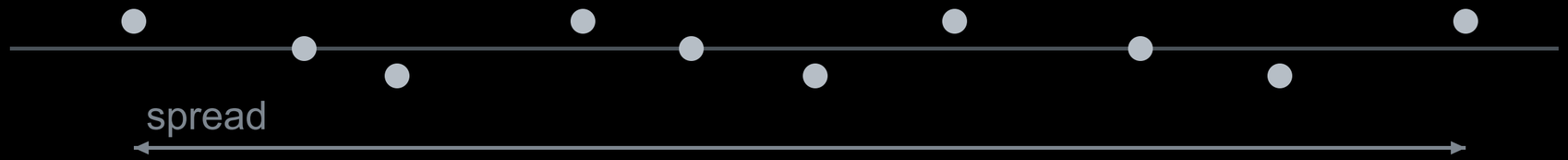


averages of five

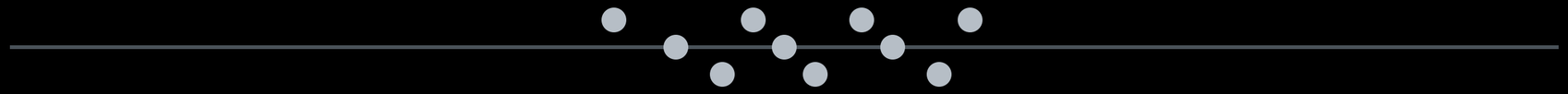


a series reveals what one reading hides

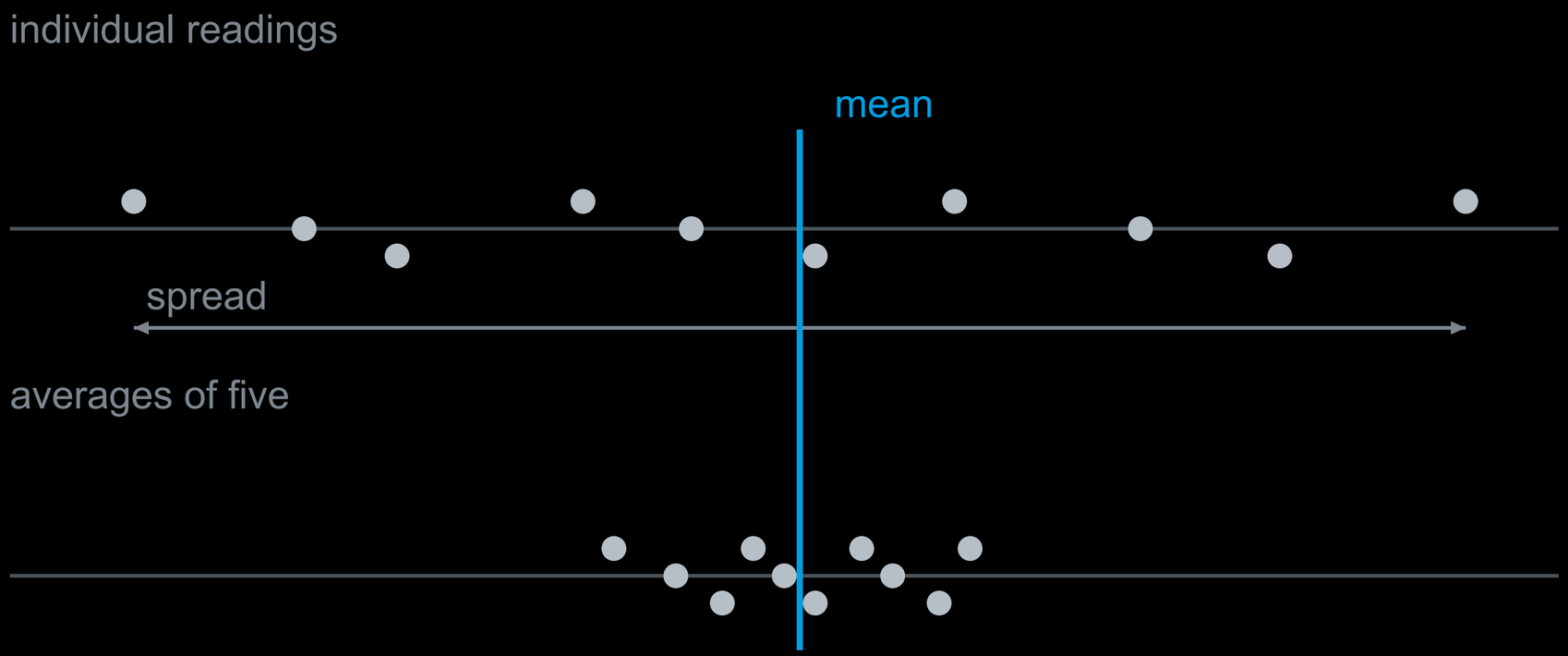
individual readings



averages of five



a series reveals what one reading hides



report the centre and the spread.

it is not the code that decides,
it is the scatter.

same sensor, same LED, same room: what differs is how far the readings move while nothing else moves.

part 2

what decides your alphabet

the same signal, averaged over longer windows

raw signal  288 readings

the same signal, averaged over longer windows

raw signal



288 readings

mean of 4



72 readings

the same signal, averaged over longer windows

raw signal



288 readings

mean of 4



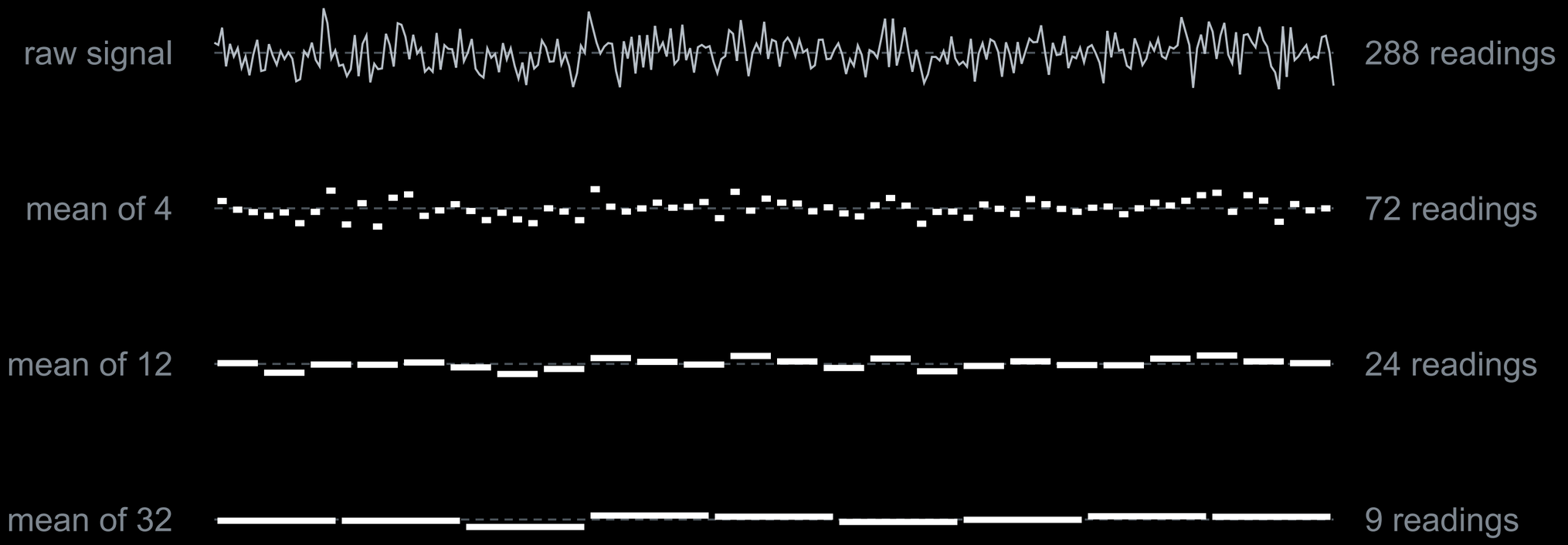
72 readings

mean of 12



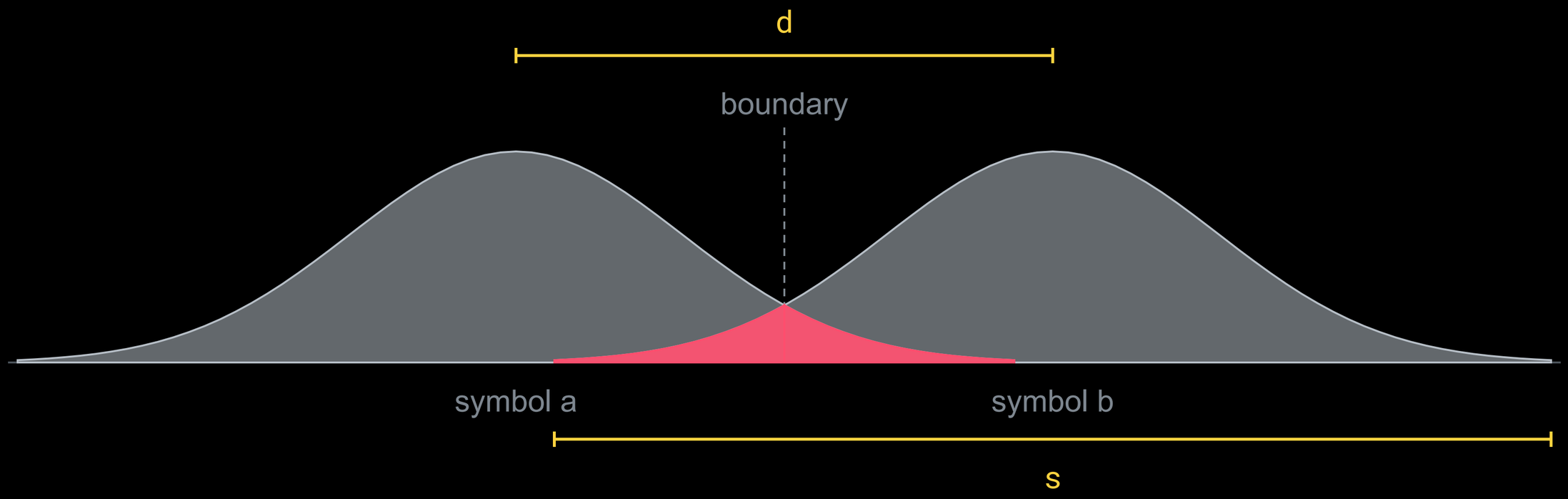
24 readings

the same signal, averaged over longer windows



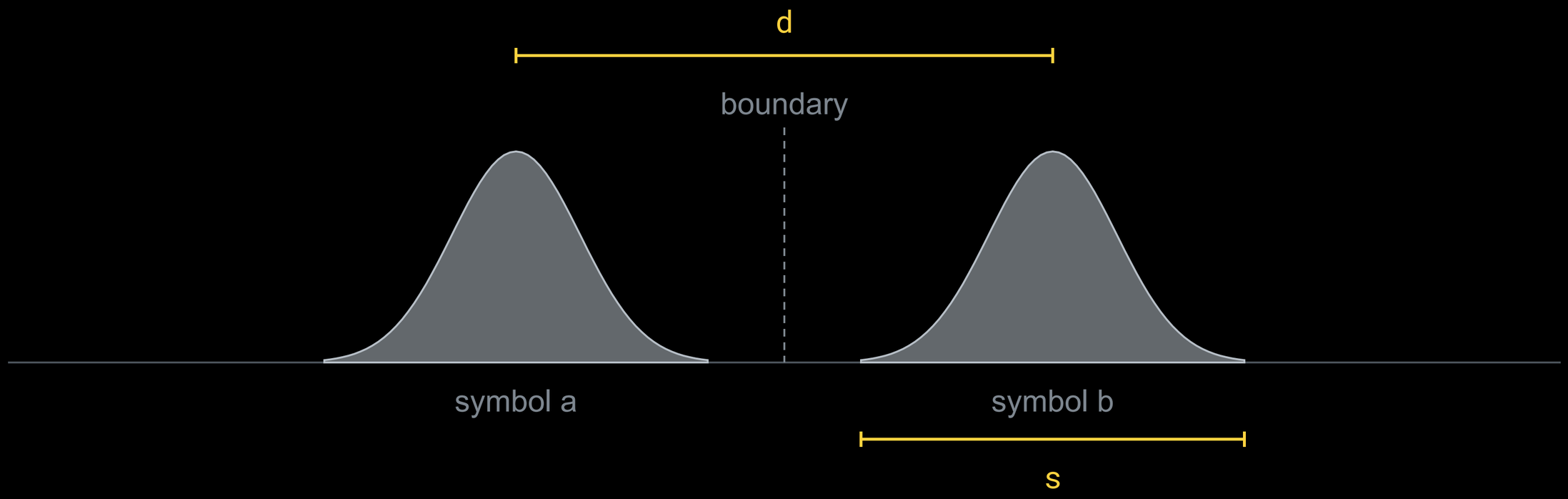
longer windows give calmer values, but fewer of them per second.

two symbols, two clouds



some readings land on the wrong side

two symbols, two clouds



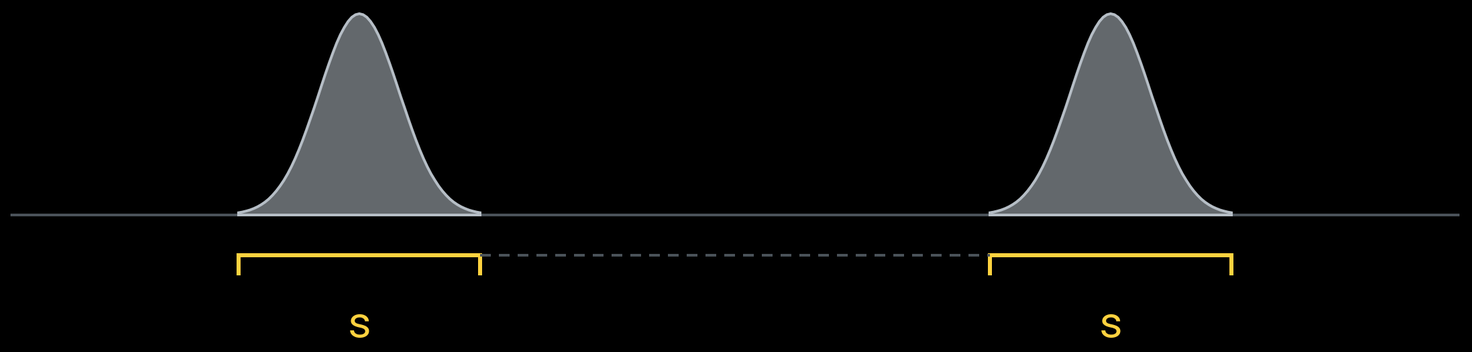
every reading lands on its own side

at the same distance, narrower clouds keep every reading on its own side.

d/s

d distance between two symbols

s spread of your readings



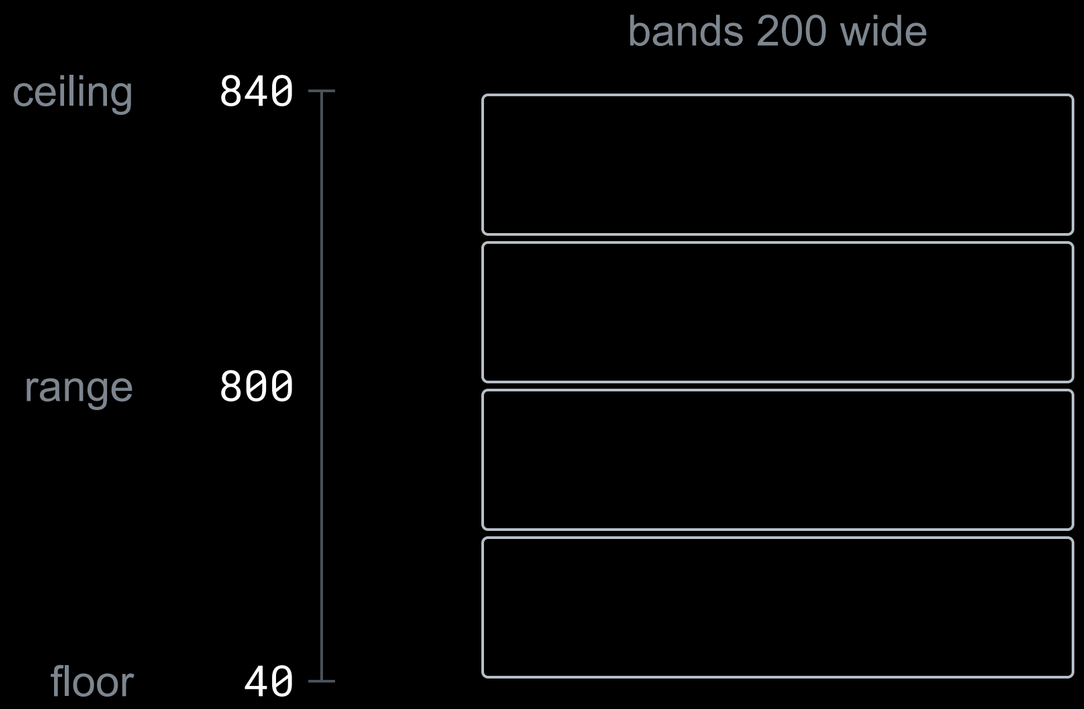
a gap between them, not a touch: that is what safely distinguishable means

your alphabet size measures nothing but this ratio.

how many symbols fit?



how many symbols fit?




```
800 / 200 = 4
```

how many symbols fit?

ceiling 840

range 800

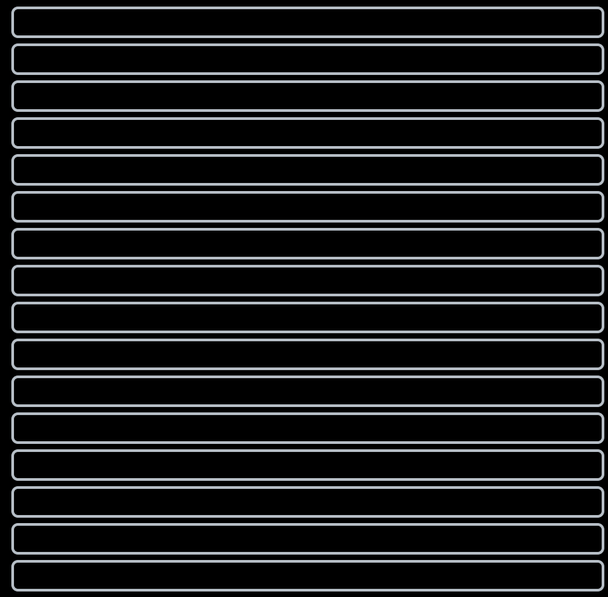
floor 40

bands 200 wide



800 / 200 = 4

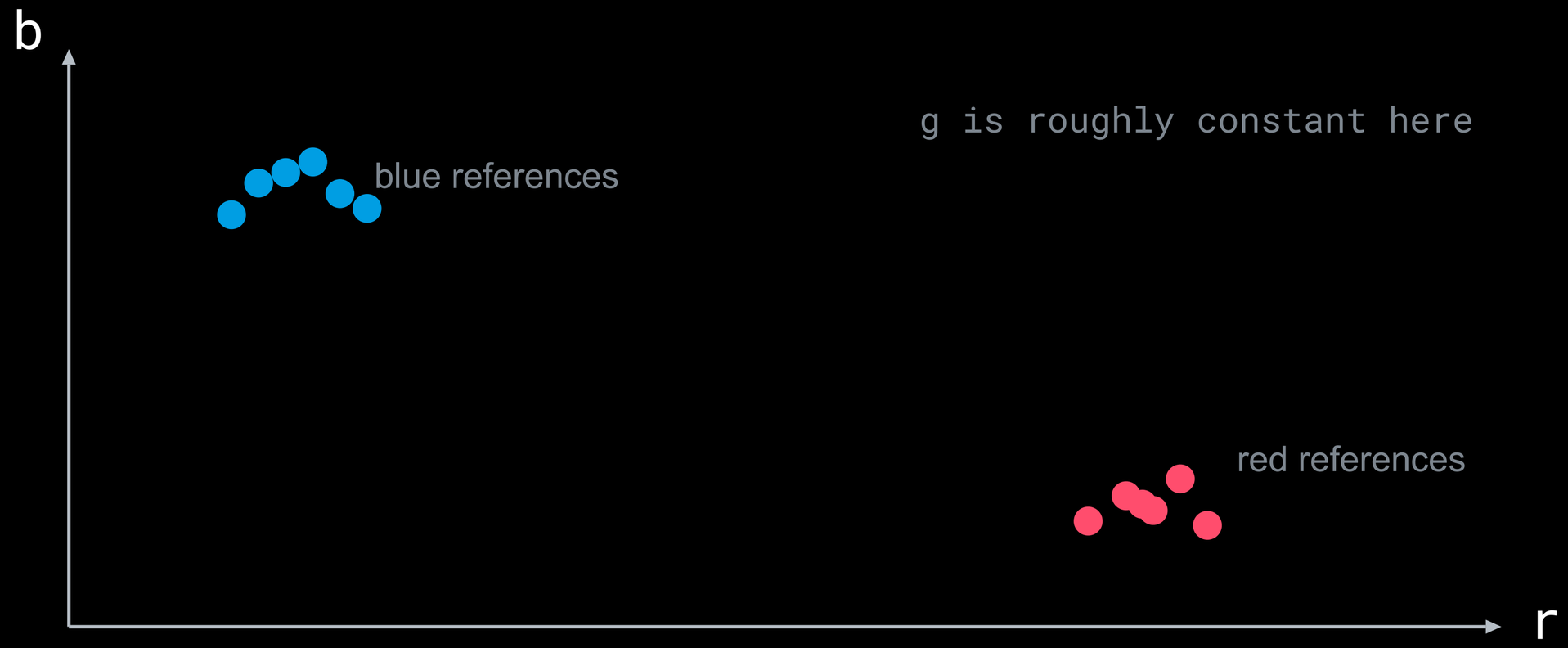
bands 50 wide



800 / 50 = 16

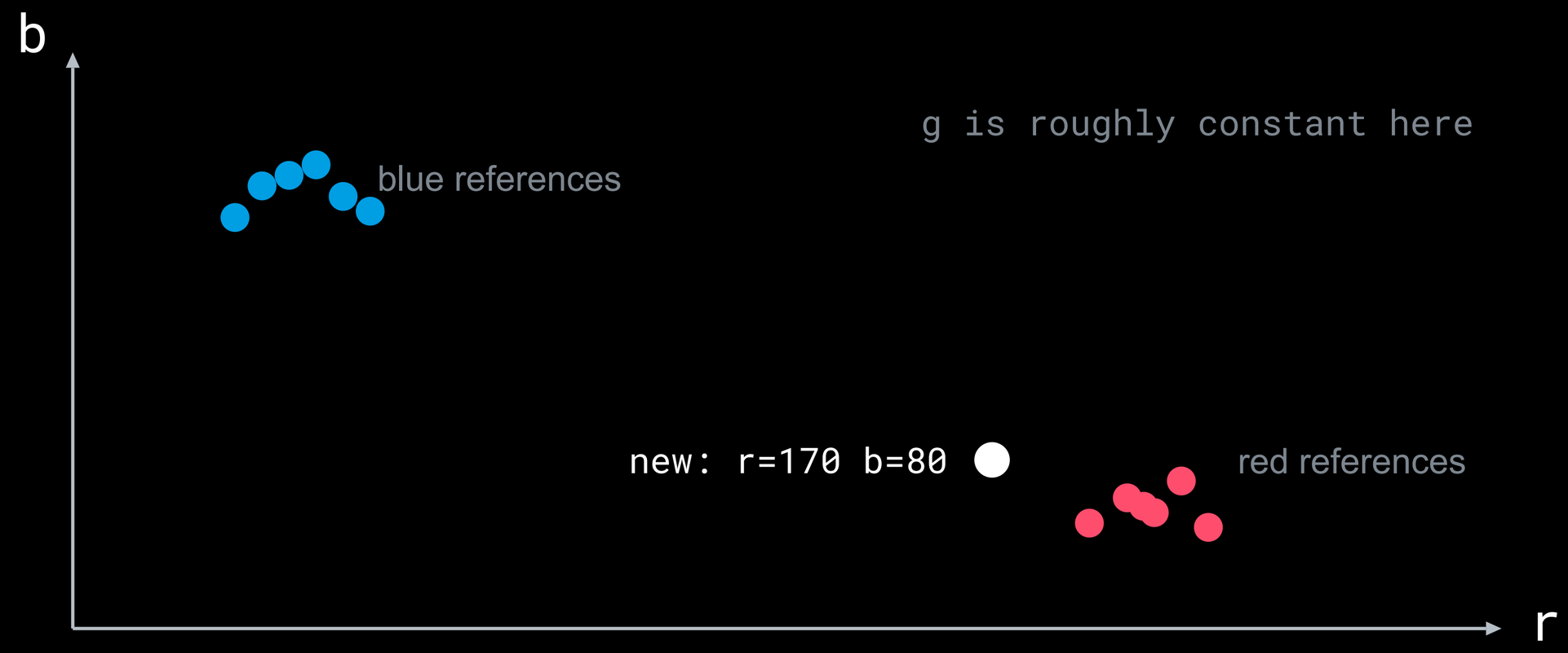
the range divided by the width of one band gives the size of your alphabet.

calibrate, then classify



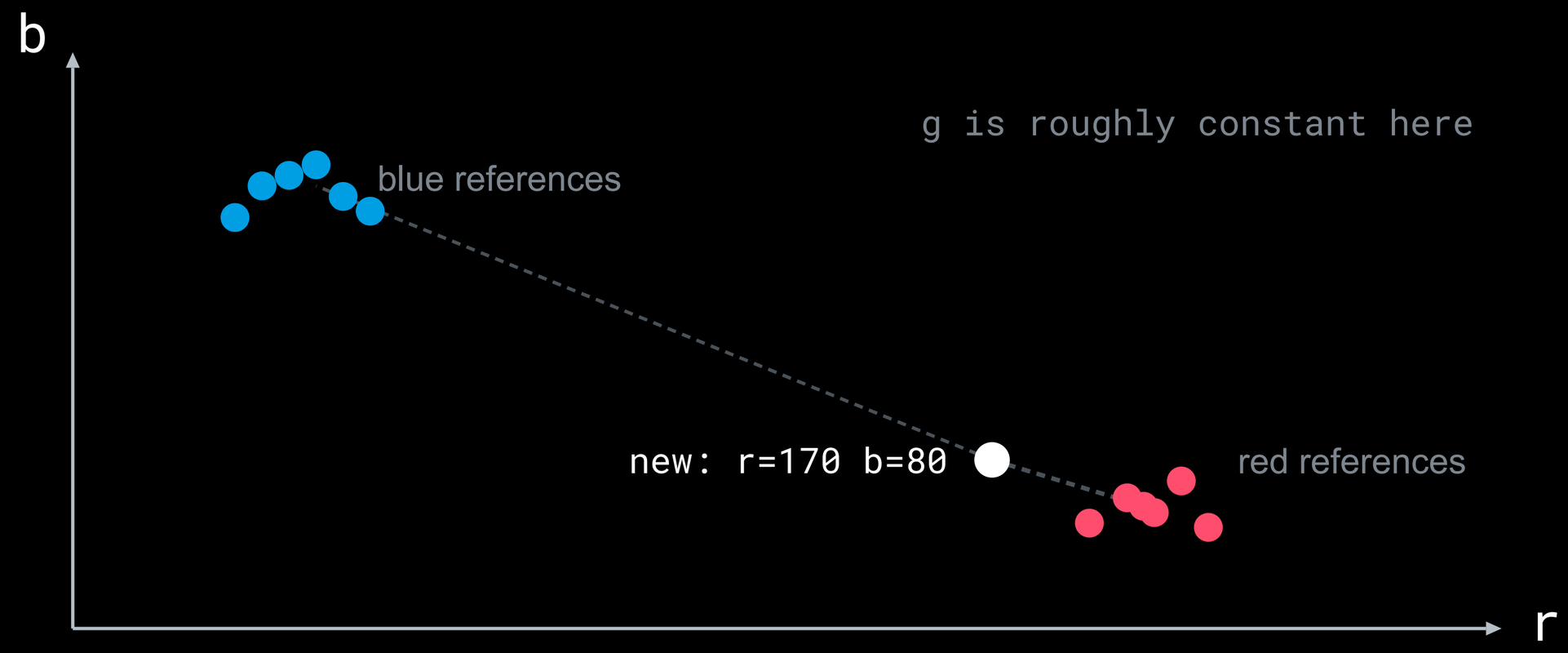
measure known colours many times

calibrate, then classify



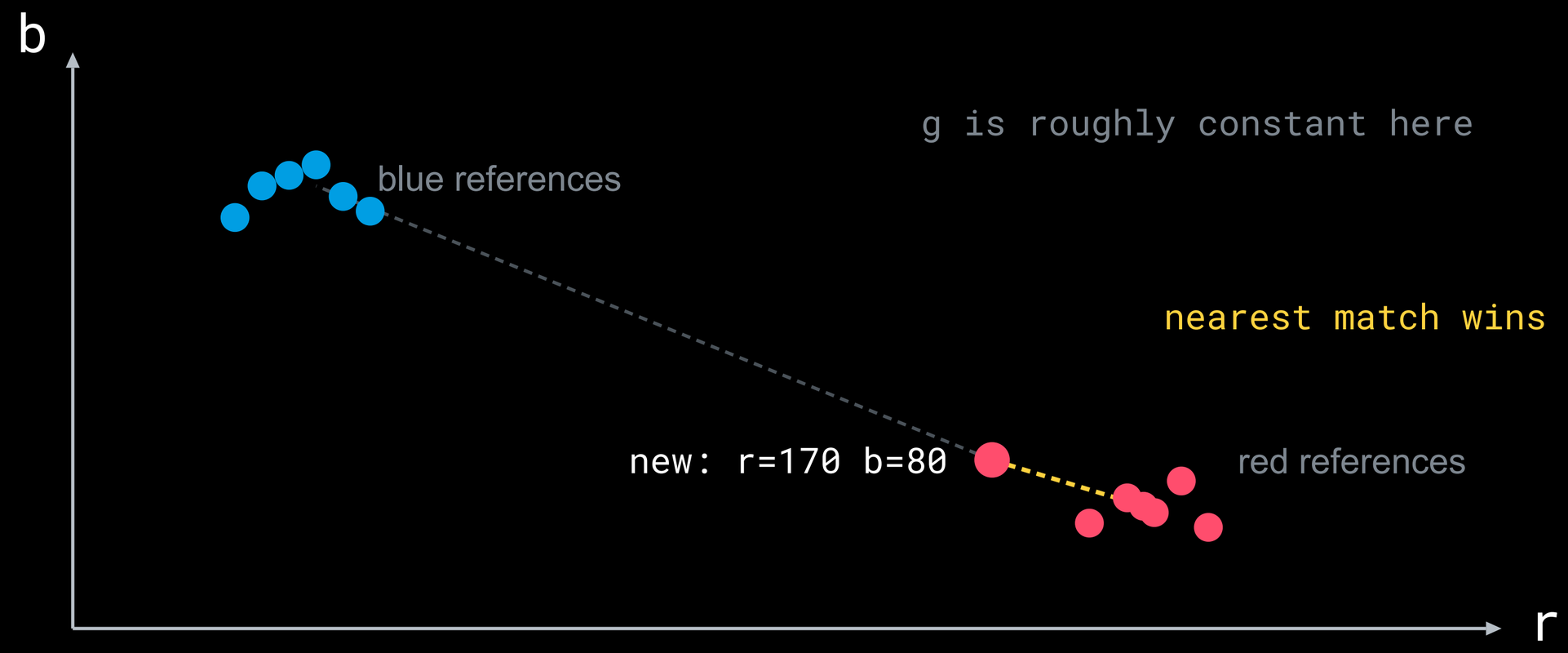
measure known colours many times

calibrate, then classify



measure known colours many times

calibrate, then classify



measure known colours many times

eight symbols, two window lengths

short window



window four times as long



a dot outside the band of its symbol is read as another symbol

simulated readings; the trade-off is the one your own link shows

what decides your alphabet

eight symbols, two window lengths

short window



6 of 30 wrong

40 symbols/s

window four times as long



a dot outside the band of its symbol is read as another symbol

simulated readings; the trade-off is the one your own link shows

what decides your alphabet

eight symbols, two window lengths

short window



6 of 30 wrong

40 symbols/s

window four times as long



0 of 30 wrong

10 symbols/s

a dot outside the band of its symbol is read as another symbol

a longer window removes the errors, but it also cuts the symbols per second.

simulated readings; the trade-off is the one your own link shows

bigger d

- **pick colours further apart**
  fewer symbols fit into the range
- **use fewer symbols**
  fewer bits per symbol
- **aim the light better**
  only craft

smaller s

- **longer integration time**
  fewer symbols per second
- **average several readings**
  fewer symbols per second
- **block ambient light**
  only craft
- **fix the geometry**
  only craft

everything else is paid for in bits per symbol or symbols per second

three of them cost nothing but craft, which is why self-built optics are allowed.

part 3

build evidence

three ways to answer the same question

it sounds right

longer readings
collect more light

three ways to answer the same question

it sounds right

longer readings
collect more light

the assistant predicts it

```
double the
integration time
```

three ways to answer the same question

it sounds right

longer readings
collect more light



the assistant predicts it

`double the integration time`



the series shows it

`41 / 50 → 47 / 50 recognised correctly`



evidence from your setup

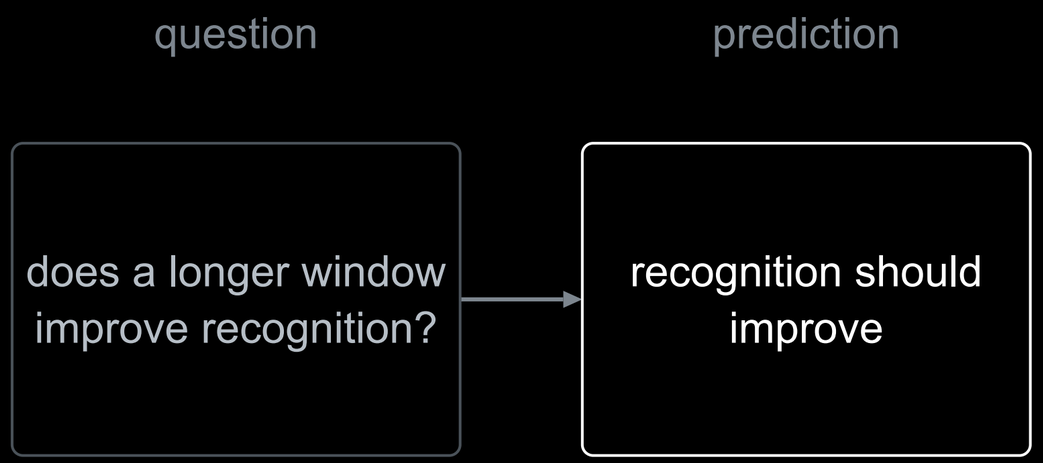
an explanation only claims something, while a measurement can decide it.

turn a claim into a test

question

does a longer window improve recognition?

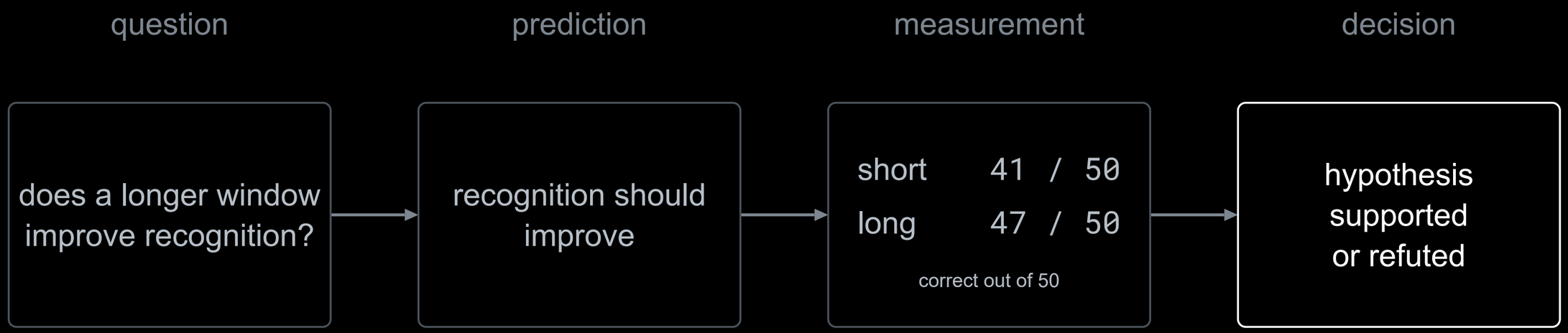
turn a claim into a test



turn a claim into a test



turn a claim into a test



write the expectation before you measure.

two teams improve their result

team a

changed distance and integration time

recognition improved

team b

changed only distance

recognition improved

two teams improve their result

team a

changed distance and integration time

recognition improved



what caused it?

team b

changed only distance

recognition improved



`distance`

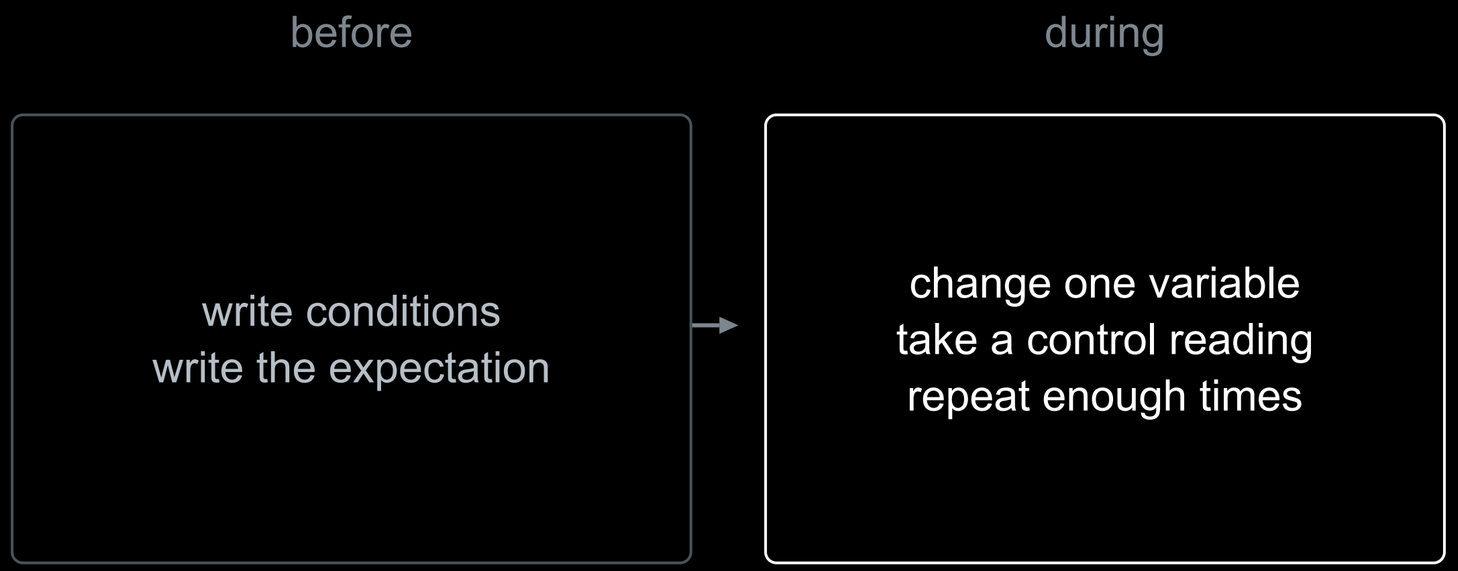
change one variable at a time.

what makes a measurement series

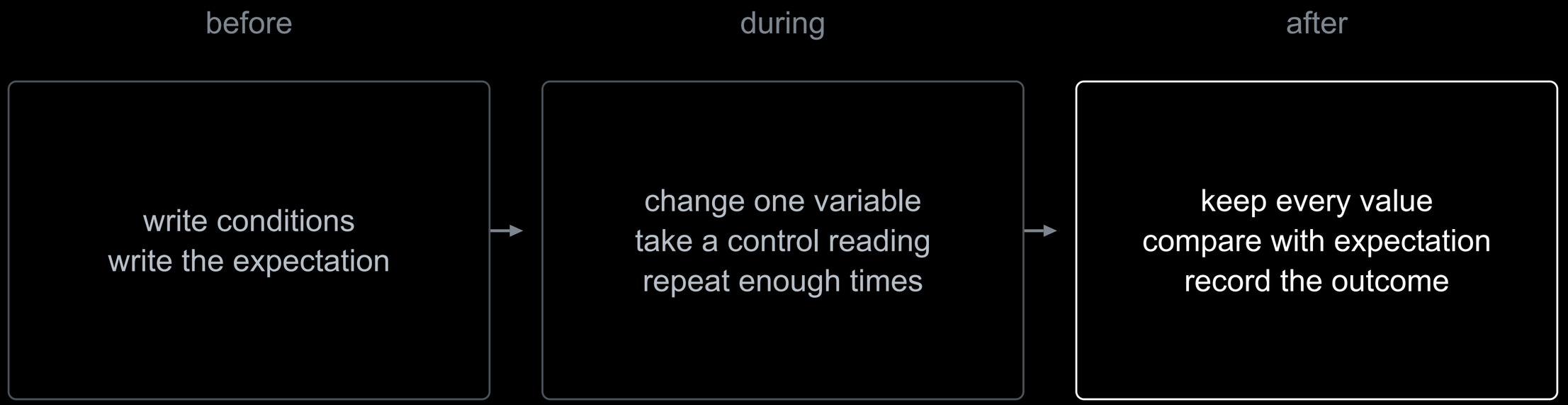
before

write conditions
write the expectation

what makes a measurement series



what makes a measurement series



a repeatable result is worth more than an impressive one.

luck or skill?





guessing succeeds about once in 32 attempts



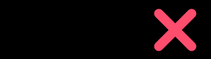
guessing succeeds about once in 32 attempts



guessing: about 1 in 1,125,899,906,842,624

luck or skill?

5 / 5



guessing succeeds about once in 32 attempts

50 / 50



`guessing: about 1 in 1,125,899,906,842,624`

evidence

when the assistant and the data disagree

assistant suggestion

"increase the gain —
recognition will improve"

your measurement series

```
gain ↑
recognition: 94% → 82%
```

when the assistant and the data disagree

assistant suggestion

"increase the gain —
recognition will improve"

your measurement series

```
gain ↑
recognition: 94% → 82%
```

✓ one variable ✓ enough repetitions ✓ control reading

when the assistant and the data disagree

assistant suggestion

"increase the gain —
recognition will improve"



your measurement series

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gain ↑
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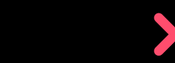

✓ one variable ✓ enough repetitions ✓ control reading

ground truth for your setup

when the assistant and the data disagree

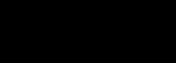
assistant suggestion

"increase the gain —
recognition will improve"



your measurement series

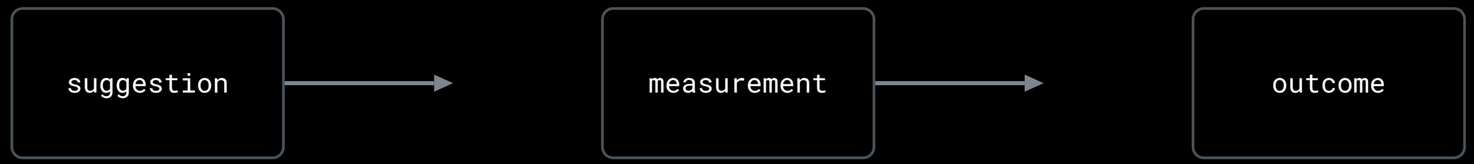
```
gain ↑
recognition: 94% → 82%
```



✓ one variable ✓ enough repetitions ✓ control reading

ground truth for your setup

log of mistakes



the fog stays, so measure how far you can see.

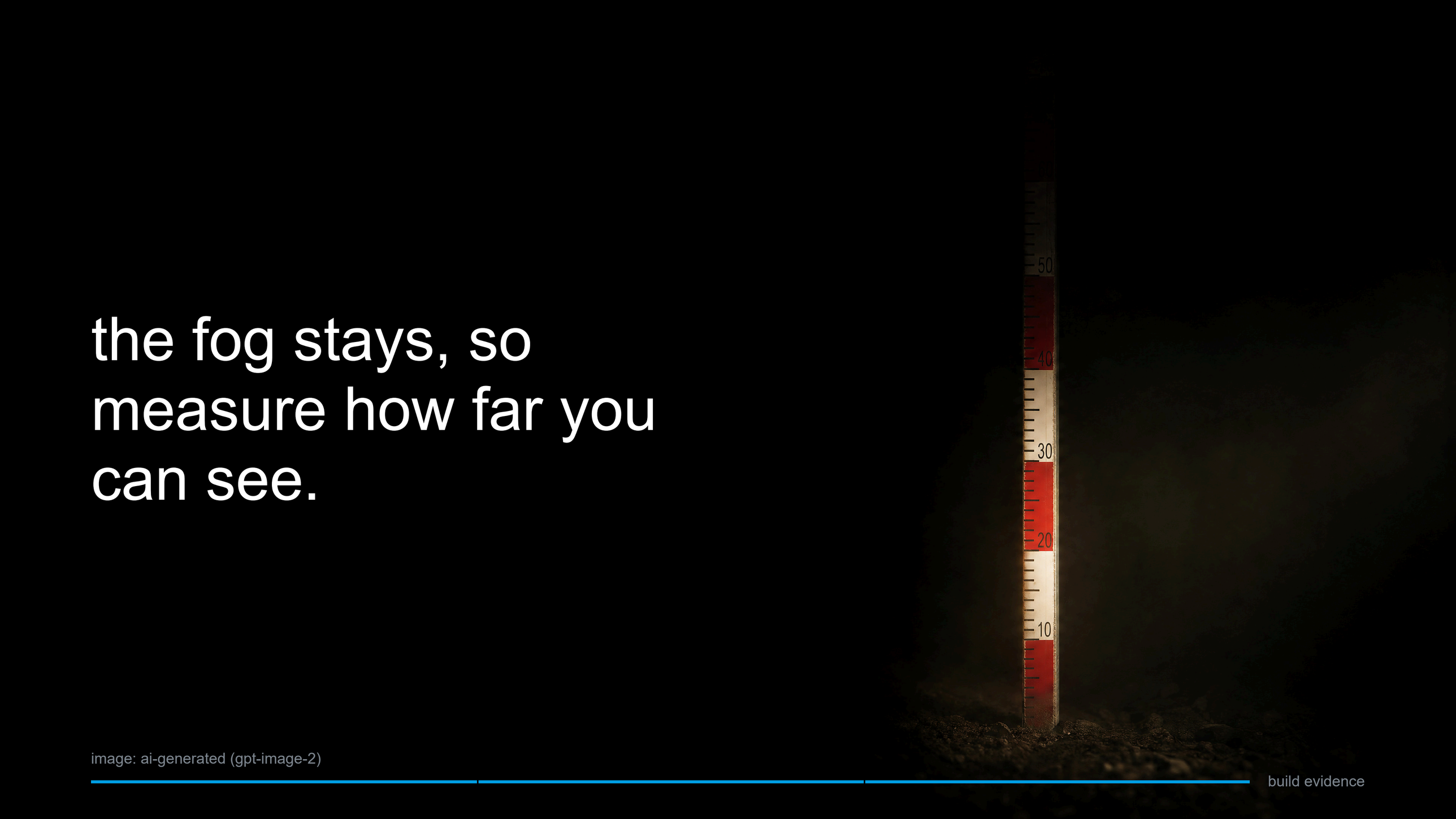


image: ai-generated (gpt-image-2)